AN UNSTABLE VARIANT OF THE MORAVA CHANGE OF RINGS THEOREM FOR $K(n)$ THEORY

WITH A COMPLETE FORMALIZATION IN LEAN 4

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Abstract. We formulate a general categorical criterion for base change of comonads in Ext and derived-functor computations via cosimplicial cobar resolutions (a generalized Shapiro's lemma for comonads). We then apply this framework to construct an unstable $\Sigma(n)$-comonad U_Σ from any base comonad U_Γ and adjunction $F \dashv U$, lift the adjunction to coalgebra categories, and prove an unstable variant of the Morava change of rings theorem for $K(n)$ theory above dimension $2p^n$. Every categorical definition, proposition, and theorem in this paper has been completely formalized and verified with zero `sorry` axioms in the Lean 4 theorem prover using Mathlib4.

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1. Introduction

The aim of the present work is twofold: to give a foundational account of comonadic base-change techniques in unstable homotopy theory, and to apply this foundation to prove an unstable variant of the celebrated Morava change of rings theorem from stable homotopy theory [8, 9], accompanied by a complete machine-checked formalization in the Lean 4 theorem prover [5, 11].

Comonads have traditionally been used to define resolving objects, and hence derived functors, in both abelian and non-abelian settings [1]. In work of Bousfield [4], a comonad G on a category $\mathcal{C}$ defines a model structure on the category of cosimplicial objects over the category $\mathcal{C}^G$ of Eilenberg–Moore coalgebras over G, in which the classical cobar resolutions $C^\bullet_G(X)$ serve as G-cofibrant replacements for computing derived mapping spaces and Ext groups.

One is naturally led to ask how morphisms of comonads interact with these cobar resolutions. Given a comonad functor $(F, \alpha) : (\mathcal{C}, G) \to (\mathcal{D}, H)$ whose underlying functor F is left adjoint to $U : \mathcal{D} \to \mathcal{C}$, the mate correspondence of Street [10] and Meyer [7] produces a comonad opfunctor $(U, \beta) : (\mathcal{D}, H) \to (\mathcal{C}, G)$ and, when $\mathcal{C}^G$ has equalizers, a right adjoint pullback functor $\alpha^* : \mathcal{D}^H \to \mathcal{C}^G$ to the pushforward $\alpha_* : \mathcal{C}^G \to \mathcal{D}^H$. Under the hypothesis that the underlying functor of α^* factors through a functor $W : \mathcal{D} \to \mathcal{C}$ on underlying categories, α^* preserves the left contractibility of cobar resolutions into cofree coalgebras, yielding our *Generalized Shapiro's Lemma for Comonads* (Proposition 2.14).

In the second part of the paper, we apply this machinery to the Morava change of rings theorem. Stably, the Morava change of rings theorem reduces an Ext computation over the Hopf algebroid (BP_*, BP_*BP) of Brown–Peterson cooperations to a computation over $(K(n)_*, \Sigma(n))$ [8]. Unstably, the category of unstable BP_*-comodules is governed by a non-additive comonad U_Γ defined via the Bendersky–Curtis–Miller excess condition [3, 2]. We show that for *any* comonad U_Γ on $\mathcal{C}$ and *any* adjunction $F \dashv U : \mathcal{D} \to \mathcal{C}$, the "sandwich" endofunctor

$$U_\Sigma := F \circ U_\Gamma \circ U : \mathcal{D} \to \mathcal{D}$$

carries a canonical comonad structure, and $\alpha := FU_\Gamma(\eta) : F \circ U_\Gamma \to U_\Sigma \circ F$ defines a comonad functor lifting $F \dashv U$ to an adjunction $\alpha_* \dashv \alpha^*$ between U_Γ-coalgebras and U_Σ-coalgebras. Specializing to $(BP_*, K(n)_*)$, we establish the unstable Miller–Ravenel decomposition for k-connected unstable comodules when $k \geq 2p^n + 1$, proving the Unstable Change of Rings Theorem (Theorem 3.10).

2. The Theory of Comonads and Generalized Shapiro's Lemma

2.1. Comonads, Eilenberg–Moore Coalgebras, and the Cokleisli Category.

We follow the conventions of Mac Lane [6], Street [10], and Meyer [7].

Definition 2.1 (Comonads and Eilenberg–Moore Coalgebras). [Lean 4: CategoryTheory.Comonad (Mathlib)] A *comonad* on a category $\mathcal{C}$ is a triple (G, ϵ, Δ) where $G : \mathcal{C} \to \mathcal{C}$ is an endofunctor and $\epsilon : G \to 1_{\mathcal{C}}$, $\Delta : G \to G^2$ are natural transformations satisfying coassociativity $\Delta_G \circ \Delta = G(\Delta) \circ \Delta$ and counitality $\epsilon_G \circ \Delta = 1_G = G(\epsilon) \circ \Delta$.

A *G-coalgebra* (X, ψ_X) consists of an object $X \in \mathcal{C}$ and a coaction $\psi_X : X \to GX$ satisfying $\Delta_X \circ \psi_X = G(\psi_X) \circ \psi_X$ and $\epsilon_X \circ \psi_X = 1_X$. Morphisms $f : (X, \psi_X) \to (Y, \psi_Y)$ in the Eilenberg–Moore category $\mathcal{C}^G$ are morphisms $f : X \to Y$ in $\mathcal{C}$ such that $G(f) \circ \psi_X = \psi_Y \circ f$.

The forgetful functor $U_G : \mathcal{C}^G \to \mathcal{C}$ is left adjoint to the cofree coalgebra functor $\mathrm{cofree}_G : \mathcal{C} \to \mathcal{C}^G$, $X \mapsto (GX, \Delta_X)$. A second category associated to G models the full subcategory of cofree coalgebras:

Definition 2.2 (Cokleisli Category). [Lean 4: CategoryTheory.Cokleisli (Mathlib)] The *Kleisli* (or *cokleisli*) category $\mathcal{C}_G$ has the same objects as $\mathcal{C}$, with hom-sets $\mathcal{C}_G(X, Y) := \mathcal{C}(GX, Y)$. The identity at X is $\epsilon_X : GX \to X$, and the composite of $f : GX \to Y$ and $g : GY \to Z$ is

$$g \circ_{\mathcal{C}_G} f := g \circ G(f) \circ \Delta_X : GX \xrightarrow{\Delta_X} G^2X \xrightarrow{Gf} GY \xrightarrow{g} Z.$$

Proposition 2.3 (Cokleisli–Coalgebra Comparison). *[Lean 4: Section2_1.cokleisliToCoalgebraFullyFait (ComonadMorphisms.lean)] There are functors $i : \mathcal{C}_G \to \mathcal{C}^G$ and $j : \mathcal{C}^G \to \mathcal{C}_G$ with $j \dashv i$, defined by:*

- *$i(X) = (GX, \Delta_X)$ and $i(f : GX \to Y) = G(f) \circ \Delta_X : GX \to GY$,*
- *$j(X, \psi_X) = X$ and $j(f : X \to Y) = f \circ \epsilon_X = \epsilon_Y \circ G(f) : GX \to Y$.*

The unit $1_{\mathcal{C}^G} \to i \circ j$ at (X, ψ_X) is ψ_X, and the counit $j \circ i \to 1_{\mathcal{C}_G}$ at Y is $\epsilon_Y = 1_Y^{\mathcal{C}_G}$.

Proof. Functoriality of i follows from coassociativity of Δ and naturality of Δ; functoriality of j follows from the right counit law $G(\epsilon_X) \circ \Delta_X = 1_{GX}$ and naturality of ϵ. The hom-set bijection $\mathcal{C}_G(j(X, \psi_X), Y) \simeq \mathcal{C}^G((X, \psi_X), i(Y))$ sends $f : GX \to Y$ to $G(f) \circ \Delta_X \circ \psi_X$ with inverse $g \mapsto \epsilon_Y \circ g$. □

2.2. Morphisms of Comonads and Induced Functors.

Definition 2.4 (Functor of Comonads). [Lean 4: Section2_2.ComonadFunctor (ComonadMorphisms.lean)] A *functor of comonads* $(F, \alpha) : (\mathcal{C}, G) \to (\mathcal{D}, H)$ consists of a functor $F : \mathcal{C} \to \mathcal{D}$ and a natural transformation $\alpha : F \circ G \to H \circ F$ such that for all $X \in \mathcal{C}$:

(1) $\Delta^H_{FX} \circ \alpha_X = H(\alpha_X) \circ \alpha_{GX} \circ F(\Delta^G_X)$,
(2) $\epsilon^H_{FX} \circ \alpha_X = F(\epsilon^G_X)$.

Proposition 2.5 (Coalgebra Pushforward)**.** *[Lean 4: Section2_2.ComonadFunctor.pushforward (ComonadMorphisms.lean)] A functor of comonads $(F, \alpha) : (\mathcal{C}, G) \to (\mathcal{D}, H)$ induces a pushforward functor $\alpha_* : \mathcal{C}^G \to \mathcal{D}^H$ commuting strictly with the forgetful functors ($U_H \circ \alpha_* = F \circ U_G$), given on objects by*

$$\alpha_*(X, \psi_X) := (FX, \alpha_X \circ F(\psi_X))$$

and on morphisms by $\alpha_*(f) := F(f)$.

Dually, with respect to Cokleisli categories and right adjoints:

Definition 2.6 (Opfunctor of Comonads)**.** [Lean 4: Section2_2.ComonadOpfunctor (ComonadMorphisms.lean)] An *opfunctor of comonads* $(U, \beta) : (\mathcal{D}, H) \to (\mathcal{C}, G)$ consists of a functor $U : \mathcal{D} \to \mathcal{C}$ and a natural transformation $\beta : G \circ U \to U \circ H$ such that for all $Y \in \mathcal{D}$:

(1) $U(\Delta^H_Y) \circ \beta_Y = \beta_{HY} \circ G(\beta_Y) \circ \Delta^G_{UY}$,
(2) $U(\epsilon^H_Y) \circ \beta_Y = \epsilon^G_{UY}$.

Proposition 2.7 (Cokleisli Pushforward)**.** *[Lean 4: Section2_2.ComonadOpfunctor.cokleisliPushforward (ComonadMorphisms.lean)] An opfunctor of comonads $(U, \beta) : (\mathcal{D}, H) \to (\mathcal{C}, G)$ induces a functor $\beta_* : \mathcal{D}_H \to \mathcal{C}_G$ given on objects by $\beta_*(Y) = UY$ and on morphisms $f \in \mathcal{D}(HY, Y')$ by $\beta_*(f) := U(f) \circ \beta_Y : GUY \to UHY \to UY'$.*

To extend β_* from cofree coalgebras to the entire Eilenberg–Moore category $\mathcal{D}^H$, recall that any H-coalgebra (Y, ψ_Y) sits in a canonical split equalizer $Y \xrightarrow{\psi_Y} HY \rightrightarrows H^2Y$. Pulling this pair across (U, β) into cofree G-coalgebras gives the parallel pair in $\mathcal{C}^G$:

$$\operatorname{cofree}_G(UY) \underset{G(\beta_Y)\circ\Delta^G_{UY}}{\overset{\operatorname{cofree}_G(U\psi_Y)}{\rightrightarrows}} \operatorname{cofree}_G(UHY).$$

Proposition 2.8 (Coalgebra Pullback via Equalizers)**.** *[Lean 4: Section2_2.ComonadOpfunctor.coalgebraP (ComonadMorphisms.lean)] Assume $\mathcal{C}^G$ has equalizers. Any opfunctor of comonads $(U, \beta) : (\mathcal{D}, H) \to (\mathcal{C}, G)$ induces a functor $\alpha^* = \beta^* : \mathcal{D}^H \to \mathcal{C}^G$ defined on objects by*

$$\alpha^*(Y, \psi_Y) := \operatorname{eq}\left(\operatorname{cofree}_G(UY) \underset{G(\beta_Y)\circ\Delta^G_{UY}}{\overset{\operatorname{cofree}_G(U\psi_Y)}{\rightrightarrows}} \operatorname{cofree}_G(UHY)\right).$$

2.3. **The Mate Correspondence and Lifted Adjunctions.** Suppose $F \dashv U : \mathcal{D} \to \mathcal{C}$ is an adjunction with unit $\sigma : 1_{\mathcal{C}} \to UF$ and counit $\tau : FU \to 1_{\mathcal{D}}$, and let G, H be comonads on $\mathcal{C}, \mathcal{D}$ respectively.

Proposition 2.9 (Mate Opfunctor and Fundamental Relation)**.** *[Lean 4: Section2_3.mateOpfunctor (ComonadMorphisms.lean)] Given a comonad functor $(F, \alpha) : (\mathcal{C}, G) \to (\mathcal{D}, H)$, define $\beta : GU \to UH$ by the mate formula*

$$\beta_Y := U(H(\tau_Y) \circ \alpha_{UY}) \circ \sigma_{GUY} : GUY \to UFGUY \to UHFUY \to UHY.$$

Then:

(1) *[Lean 4: Section2_3.mate_fundamental_rel_counit (ComonadMorphisms.lean)]* *β satisfies the fundamental mate identities:*

$$\tau_{HY} \circ F(\beta_Y) = H(\tau_Y) \circ \alpha_{UY} \quad \text{and} \quad \beta_{FX} \circ G(\sigma_X) = U(\alpha_X) \circ \sigma_{GX}.$$

(2) *(U, β) is an opfunctor of comonads $(\mathcal{D}, H) \to (\mathcal{C}, G)$.*

Proposition 2.10 (Lifted Adjunction on Coalgebras). *[Lean 4: Section2_3.liftedAdjunction (ComonadMorphisms.lean)] Suppose $F \dashv U$, $(F, \alpha) : (\mathcal{C}, G) \to (\mathcal{D}, H)$ is a functor of comonads, and $\mathcal{C}^G$ has equalizers. Then the pushforward $\alpha_* : \mathcal{C}^G \to \mathcal{D}^H$ is left adjoint to the pullback $\alpha^* : \mathcal{D}^H \to \mathcal{C}^G$:*

$$\alpha_* \dashv \alpha^* : \mathcal{D}^H \to \mathcal{C}^G.$$

For $(X, \psi_X) \in \mathcal{C}^G$, the lifted unit $\tilde{\sigma}_X : X \to \alpha^*\alpha_* X$ is the unique morphism into the equalizer induced by $G(\sigma_X) \circ \psi_X : X \to GUFX$. For $(Y, \psi_Y) \in \mathcal{D}^H$, the lifted counit $\tilde{\tau}_Y : \alpha_*\alpha^* Y \to Y$ has underlying morphism $\tau_Y \circ F(\epsilon^G_{UY} \circ \iota_Y) : F(\alpha^* Y) \to FGUY \to FUY \to Y$.

2.4. **The Cobar Resolution and Contractibility.** Let G be a comonad on $\mathcal{C}$. The adjunction $U_G \dashv \mathrm{cofree}_G$ induces a monad (T, μ, η) on $\mathcal{C}^G$ with underlying functor $T(X, \psi_X) = (GX, \Delta_X)$, unit $\eta_X = \psi_X : X \to TX$, and multiplication $\mu_X = G(\epsilon_X) : T^2 X \to TX$.

Definition 2.11 (Cosimplicial Cobar Resolution). [Lean 4: Section2_4.cobarObj (CobarResolution.lean)] For $X \in \mathcal{C}^G$, the *cobar complex* $C^\bullet_G(X)$ has n-th cosimplicial degree $C^n_G(X) := T^{n+1}(X) = G^{n+1}(X)$, coaugmentation $d^0_{-1} = \eta_X = \psi_X : X \to TX$, cofaces $d^i_n : T^{n+1}X \to T^{n+2}X$ $(0 \le i \le n+1)$, and codegeneracies $s^j_n : T^{n+2}X \to T^{n+1}X$ $(0 \le j \le n)$ defined recursively by applying T^i to η and μ.

Proposition 2.12 (Cosimplicial Identities and Left Contractibility). *[Lean 4: Section2_4.cobarHomLeftContractibleTower (CobarResolution.lean)] For every $X \in \mathcal{C}^G$:*

(1) *[Lean 4: Section2_4.cobarCoface_comp_cobarCoface (CobarResolution.lean)] $(C^\bullet_G(X), d^i_n, s^j_n)$ satisfies all five cosimplicial identities and $d^0_0 \circ d^0_{-1} = d^1_0 \circ d^0_{-1}$.*

(2) *For every cofree G-coalgebra $G(J) = \mathrm{cofree}_G(J)$, the augmented simplicial hom-tower*

$$\mathrm{Hom}_{\mathcal{C}^G}(C^\bullet_G(X), G(J)) \longrightarrow \mathrm{Hom}_{\mathcal{C}^G}(X, G(J))$$

admits an explicit left contraction $h_{-1}(f) := G(f) \circ \Delta_X$, $h_n(f) := G(f) \circ \Delta_{G^{n+1}X}$, satisfying $d_0 \circ h_{-1} = \mathrm{id}$, $d_0 \circ h_n = \mathrm{id}$, $d_{i+1} \circ h_n = h_{n-1} \circ d_i$, and $s_0 \circ \mathrm{aug} = h_{-1}$. Consequently, $X \to C^\bullet_G(X)$ is a weak G-resolution in the sense of Bousfield [4].

2.5. **A Generalized Shapiro's Lemma for Comonads.**

Proposition 2.13 (Naturality Triangle for the Lifted Adjunction)**.** *[Lean 4: Section2_5.shapiro_naturality_triangle_coaug (CobarResolution.lean)] Under the hypotheses of Proposition 2.10, for any $M \in \mathcal{C}^G$, $N \in \mathcal{D}^H$, and morphism $f : M \to \alpha^* N$, the naturality of the lifted adjunction $\tilde{\gamma} : \mathrm{Hom}_{\mathcal{C}^G}(M, \alpha^* -) \simeq \mathrm{Hom}_{\mathcal{D}^H}(\alpha_* M, -)$ along the cobar coaugmentation $\psi_N : N \to H(N)$ yields the commutative triangle*

$$\tilde{\gamma}_{M,H(N)}(f \circ_{post} \alpha^*(\psi_N)) = \tilde{\gamma}_{M,N}(f) \circ \psi_N.$$

Proposition 2.14 (Generalized Shapiro's Lemma for Comonads)**.** *[Lean 4: Section2_5.generalizedShapiroCobarIso (CobarResolution.lean)] Suppose $F \dashv U : \mathcal{D} \to \mathcal{C}$, $(F, \alpha) : (\mathcal{C}, G) \to (\mathcal{D}, H)$ is a functor of comonads, and $\mathcal{C}^G$ has equalizers. Suppose further that there is a functor $W : \mathcal{D} \to \mathcal{C}$ and a natural isomorphism $U_G \circ \alpha^* \cong W \circ U_H$. Then for every $M \in \mathcal{C}^G$ and $N \in \mathcal{D}^H$:*

(1) *[Lean 4: Section2_5.pullbackCobarCofreeLeftContractibleTower (CobarResolution.lean)] For every $J \in \mathcal{C}$, the augmented simplicial hom-tower $\mathrm{Hom}_{\mathcal{C}^G}(\alpha^*(C_H^\bullet(N)), G(J)) \to \mathrm{Hom}_{\mathcal{C}^G}(\alpha^* N, G(J))$ is isomorphic to $\mathrm{Hom}_{\mathcal{C}}(W(C_H^\bullet(N)), J) \to \mathrm{Hom}_{\mathcal{C}}(W(N), J)$, which is left-contractible via $W(\epsilon^H)^*$. Thus $\alpha^* N \to \alpha^*(C_H^\bullet(N))$ is a Bousfield weak G-resolution.*

(2) *[Lean 4: Section2_5.generalizedShapiroExtCobarIso (CobarResolution.lean)] The lifted adjunction $\alpha_* \dashv \alpha^*$ induces an isomorphism of coaugmented cosimplicial hom-towers:*

$$\mathrm{Hom}_{\mathcal{C}^G}\big(M,\, \alpha^*(C_H^\bullet(N))\big) \;\cong\; \mathrm{Hom}_{\mathcal{D}^H}\big(\alpha_* M,\, C_H^\bullet(N)\big),$$

and hence an isomorphism of comonadic Ext *groups* $\mathrm{Ext}_G(M, \alpha^* N) \cong \mathrm{Ext}_H(\alpha_* M, N)$.

3. An Unstable Morava Change of Rings Theorem

3.1. **The Canonical Split Equalizer of a Coalgebra and the Stable Case.**

Proposition 3.1 (Canonical Equalizer Presentation of Coalgebras)**.** *[Lean 4: Section3_1.coalgebraCanonicalEqualizerIso (UnstableChangeOfRings.lean)] For any comonad G on $\mathcal{C}$ and any G-coalgebra $X = (X, \psi_X) \in \mathcal{C}^G$, the coaction $\psi_X : X \to \mathrm{cofree}_G(X)$ exhibits X as the limit (equalizer) in $\mathcal{C}^G$ of the canonical parallel pair*

$$X \xrightarrow{\psi_X} \mathrm{cofree}_G(X) \underset{\Delta_X}{\overset{\mathrm{cofree}_G(\psi_X)}{\rightrightarrows}} \mathrm{cofree}_G(GX).$$

In the classical stable setting [8, 9], the projection of Hopf algebroids $\pi : (BP_*, BP_*BP) \to (K(n)_*, \Sigma(n))$ induces a comonad functor on module categories, and Miller–Ravenel's isomorphism $BP_*K(n) \cong B(n)_* \otimes_{K(n)_*} \Sigma(n)$

identifies the cotensor $\pi^*N = BP_*K(n)\square_{\Sigma(n)}N \cong B(n)_* \otimes_{K(n)_*} N$. Proposition 2.14 then immediately recovers the stable Morava Change of Rings isomorphism ([Lean 4: Section3_1.stableChangeOfRingsCobarIso (UnstableChangeOfRings.lean)]).

3.2. **The General Unstable Sandwich Comonad U_Σ.** In the unstable setting, the comonad U_Γ on BP_*-modules is defined on free modules by the excess bound $2\,\mathrm{len}(I) < \deg(m)$ [3, 2] and extended to (-1)-connected modules via free presentations; in particular, U_Γ is not given by tensoring with a coalgebra. Nevertheless, our formalization reveals that the construction of the unstable target comonad U_Σ and the comonad functor $\alpha : FU_\Gamma \to U_\Sigma F$ holds in complete categorical generality for *any* comonad U_Γ on $\mathcal{C}$ and *any* adjunction $F \dashv U : \mathcal{D} \to \mathcal{C}$ (without requiring the counit $FU \to 1_\mathcal{D}$ to be an isomorphism).

Definition 3.2 (Unstable Sandwich Comonad U_Σ)**.** [Lean 4: Section3_2.unstableSigmaComonad (UnstableChangeOfRings.lean)] Let U_Γ be a comonad on $\mathcal{C}$ and $F \dashv U : \mathcal{D} \to \mathcal{C}$ an adjunction with unit $\eta : 1_\mathcal{C} \to UF$ and counit $\varepsilon : FU \to 1_\mathcal{D}$. Define the endofunctor $U_\Sigma : \mathcal{D} \to \mathcal{D}$ by

$$U_\Sigma(N) := F(U_\Gamma(U(N))),$$

with counit $\epsilon^\Sigma_N := \varepsilon_N \circ F(\epsilon^\Gamma_{UN}) : FU_\Gamma UN \to FUN \to N$ and comultiplication

$$\Delta^\Sigma_N := F(U_\Gamma(\eta_{U_\Gamma UN})) \circ F(\Delta^\Gamma_{UN}) : FU_\Gamma UN \to FU^2_\Gamma UN \to FU_\Gamma UFU_\Gamma UN.$$

Theorem 3.3 (U_Σ is a Comonad)**.** *[Lean 4: Section3_2.unstableSigmaComonad (UnstableChangeOfRings.lean)] For any comonad U_Γ on $\mathcal{C}$ and any adjunction $F \dashv U : \mathcal{D} \to \mathcal{C}$, $(U_\Sigma, \epsilon^\Sigma, \Delta^\Sigma)$ is a comonad on $\mathcal{D}$.*

Proof. Coassociativity $\Delta^\Sigma_{U_\Sigma N} \circ \Delta^\Sigma_N = U_\Sigma(\Delta^\Sigma_N) \circ \Delta^\Sigma_N$ follows by applying F to the composition of: (1) naturality of Δ^Γ along $\eta_{U_\Gamma UN}$, (2) coassociativity of Δ^Γ at UN, and (3) naturality of η along Δ^Γ_{UN}. Left counitality follows from the left adjunction triangle $\varepsilon_{FX} \circ F(\eta_X) = 1_{FX}$ combined with the left counit law for U_Γ; right counitality follows from naturality of ε, the right adjunction triangle $U(\varepsilon_N) \circ \eta_{UN} = 1_{UN}$, and the right counit law for U_Γ. □

Definition 3.4 (Unstable Base-Change Comonad Morphism)**.** [Lean 4: Section3_2.unstableComonadFunct (UnstableChangeOfRings.lean)] Define the natural transformation $\alpha : F \circ U_\Gamma \to U_\Sigma \circ F$ at $M \in \mathcal{C}$ by

$$\alpha_M := F(U_\Gamma(\eta_M)) : F(U_\Gamma(M)) \longrightarrow F(U_\Gamma(U(F(M)))) = U_\Sigma(F(M)).$$

Theorem 3.5 ((F, α) is a Functor of Comonads)**.** *[Lean 4: Section3_2.unstableComonadFunctor (UnstableChangeOfRings.lean)] The pair $(F, \alpha) : (\mathcal{C}, U_\Gamma) \to (\mathcal{D}, U_\Sigma)$ is a functor of comonads.*

Proof. Compatibility with comultiplication reduces inside $F(-)$ to naturality of Δ^Γ along η_M and naturality of η along Δ^Γ_M. Compatibility with the counits reduces to naturality of ϵ^Γ along η_M followed by the triangle identity $\varepsilon_{FM} \circ F(\eta_M) = 1_{FM}$. □

3.3. The Unstable Cotensor and Lifted Adjunction.

Theorem 3.6 (Unstable Lifted Adjunction and Explicit Mate). *[Lean 4: Section3_2.unstableLiftedAdjunction (UnstableChangeOfRings.lean)] Suppose $\mathcal{C}^{U_\Gamma}$ has equalizers.*

(1) *[Lean 4: Section2_3.unstableMateNatTrans_app (UnstableChangeOfRings.lean)] When the counit $\varepsilon : FU \to 1_{\mathcal{D}}$ satisfies $\varepsilon_{FX} = 1_{FX}$ (or more generally, up to the canonical counit isomorphism), the mate opfunctor $(U, \beta) : (\mathcal{D}, U_\Sigma) \to (\mathcal{C}, U_\Gamma)$ has underlying natural transformation*

$$\beta_N = \eta_{U_\Gamma(UN)} : U_\Gamma(UN) \longrightarrow U(F(U_\Gamma(UN))) = U(U_\Sigma(N)).$$

(2) *The comonad functor (F, α) induces an adjoint pair of coalgebra functors*

$$\alpha_* : \mathcal{C}^{U_\Gamma} \longleftrightarrow \mathcal{D}^{U_\Sigma} : \alpha^*.$$

Theorem 3.7 (Lifted Counit Isomorphism). *[Lean 4: Section3_2.unstablePushforwardCotensorIso (UnstableChangeOfRings.lean)] Suppose $\mathcal{C}^{U_\Gamma}$ has equalizers, the functor $F : \mathcal{C} \to \mathcal{D}$ preserves the defining equalizer of $\alpha^* N$, and the counit $\varepsilon : FU \to 1_{\mathcal{D}}$ is an isomorphism. Then for every $N \in \mathcal{D}^{U_\Sigma}$, the lifted counit*

$$\tilde{\varepsilon}_N : \alpha_*(\alpha^* N) \xrightarrow{\sim} N$$

is an isomorphism of U_Σ-coalgebras ([Lean 4: Section3_2.isIso_unstableLiftedCounit (UnstableChangeOfRings.lean)]).

Proof. A key structural observation verified by reflexivity (`rfl`) in Lean is that for any $Y \in \mathcal{D}$, pushing forward the cofree U_Γ-coalgebra on UY along α_* yields the cofree U_Σ-coalgebra on Y on the nose:

$$\alpha_*\big(\operatorname{cofree}_{U_\Gamma}(UY)\big) = \operatorname{cofree}_{U_\Sigma}(Y) \quad \text{[Lean 4: Section3_2.unstablePushforward_obj_cofree (UnstableChange}$$

Moreover, applying α_* to the top and bottom arrows of the equalizer diagram defining $\alpha^* N$ yields $\operatorname{cofree}_{U_\Sigma}(\varepsilon_N)^{-1} \circ \operatorname{cofree}_{U_\Sigma}(\psi_N)$ and $\operatorname{cofree}_{U_\Sigma}(\varepsilon_N)^{-1} \circ \Delta^\Sigma_N$, respectively. Since F preserves this equalizer and $\operatorname{cofree}_{U_\Sigma}(\varepsilon_N)$ is an isomorphism, $\alpha_*(\alpha^* N)$ is the equalizer in $\mathcal{D}^{U_\Sigma}$ of $(\operatorname{cofree}_{U_\Sigma}(\psi_N), \Delta^\Sigma_N)$. By Proposition 3.1, N is also the equalizer of this pair, and the unique comparison isomorphism is precisely the lifted counit $\tilde{\varepsilon}_N$. □

3.4. Connectivity, Height-n Comodules, and the Unstable Change of Rings.

In the topological setting where $(A, \Gamma) = (BP_*, BP_*BP)$ and $B = K(n)_* = \mathbb{F}_p[v_n, v_n^{-1}]$, we examine the unstable excess conditions needed to destabilize the Miller–Ravenel isomorphism:

Proposition 3.8 (Unstable Excess Bounds and Left/Right v_n^{-1} Inversion). *[Lean 4: Section3_2.excess_ul_action_iff (UnstableChangeOfRings.lean)]*

(1) *In BP_*BP, the formal group law relation $\eta_R(v_{n+l} \otimes \iota_k) \equiv (h_l^{p^n} \cdot v_n - v_n^{p^l} h_l) \otimes \iota_k$ has $h_l^{p^n} \otimes v_n \iota_k$ of length p^n and degree $k + 2p^n - 2$, satisfying the excess inequality $2p^n < k + 2p^n - 2$ if and only if $k \geq 3$ ([Lean 4: Section3_2.excess_vn_shift_iff (UnstableChangeOfRings.lean)]).*

(2) *Dividing by v_n to define the $b(n)_*\{\iota_k\}$-action $u_l \otimes \iota_k \mapsto (h_l^{p^n} - v_n^{p^l-1}h_l)\otimes\iota_k$ requires $2p^n < k$, i.e., $k \geq 2p^n+1$ (*[Lean 4: `Section3_2.excess_ul_action_iff (UnstableChangeOfRings.lean)`]*).*

(3) *[Lean 4: `Section3_2.prop3_2_2_left_right_inv_comm (UnstableChangeOfRings.lean)`] In this range ($k \geq 2p^n+1$), the left and right actions of v_n agree, which implies that the left and right actions of v_n^{-1} agree and commute with the conjugation twist $c\circ T$. Since U_Γ and U_Σ commute with colimits, for any k-connected U_Γ-comodule M with $k \geq 2p^n+1$ we obtain a natural isomorphism (*[Lean 4: `Section3_2.prop3_2_2_colimit_iso (UnstableChangeOfRings.lean)`]*)*

$$U_\Gamma(K(n)_* \otimes_{BP_*} M) \;\cong\; B(n)_* \otimes_{K(n)_*} U_\Sigma(K(n)_* \otimes_{BP_*} M).$$

Proposition 3.9 (Unstable Cotensor Identification and Height-n Unit Isomorphism)**.** *[Lean 4: `Section3_2.prop3_2_3_unitPullbackIso (UnstableChangeOfRings.lean)`] Let $B(n)_* \otimes_{K(n)_*} - : \mathcal{D} \to \mathcal{C}$ preserve the canonical equalizer of U_Σ-coalgebras.*

(1) *[Lean 4: `Section3_2.prop3_2_4_cotensor_cofree (UnstableChangeOfRings.lean)`] For any cofree U_Σ-comodule $U_\Sigma(Y)$, $\alpha^*(U_\Sigma(Y)) \cong U_\Gamma(U(Y))$.*

(2) *[Lean 4: `Section3_2.prop3_2_3_unitPullbackIso (UnstableChangeOfRings.lean)`] For any k-connected U_Γ-comodule M with $k \geq 2p^n+1$, the underlying object of $\alpha^*\alpha_*M$ is naturally isomorphic to $B(n)_* \otimes_{BP_*} M$.*

(3) *[Lean 4: `Section3_2.liftedUnitIsoOfHeightNConnected (UnstableChangeOfRings.lean)`] Consequently, if M is of height n and connectivity $k \geq 2p^n + 1$, the lifted adjunction unit $\tilde{\sigma}_M : M \xrightarrow{\sim} \alpha^*\alpha_*M$ is an isomorphism of U_Γ-coalgebras.*

Theorem 3.10 (Unstable Morava Change of Rings)**.** *[Lean 4: `Section3_2.unstableChangeOfRingsCobarIso (UnstableChangeOfRings.lean)`] Let M be a height-n unstable U_Γ-comodule of connectivity $k \geq 2p^n + 1$. Then for any test coalgebra $P \in \mathcal{C}^{U_\Gamma}$ (such as BP_*), there is a canonical isomorphism of coaugmented cosimplicial cobar hom-towers*

$$\mathrm{Hom}_{\mathcal{C}^{U_\Gamma}}\big(P,\, C^\bullet_{U_\Gamma}(M)\big) \;\cong\; \mathrm{Hom}_{\mathcal{D}^{U_\Sigma}}\big(\alpha_*P,\, C^\bullet_{U_\Sigma}(\alpha_*M)\big),$$

and the pulled-back cobar complex $\alpha^*(C^\bullet_{U_\Sigma}(\alpha_*M))$ is a left-contractible Bousfield weak U_Γ-resolution ([Lean 4: `Section3_2.unstablePullbackCobarLeftContractibleTower (UnstableChangeOfRings.lean)`]*). In particular,*

$$\mathrm{Ext}_{U_\Gamma}(BP_*,\, M) \;\cong\; \mathrm{Ext}_{U_\Sigma}(K(n)_*,\, K(n)_* \otimes_{BP_*} M).$$

Proof. By Proposition 3.9(3), the lifted unit $\tilde{\sigma}_M : M \xrightarrow{\sim} \alpha^*\alpha_*M$ is an isomorphism of U_Γ-coalgebras, inducing an isomorphism of cobar hom-towers $\mathrm{Hom}_{\mathcal{C}^{U_\Gamma}}(P, C^\bullet_{U_\Gamma}(M)) \cong \mathrm{Hom}_{\mathcal{C}^{U_\Gamma}}(P, C^\bullet_{U_\Gamma}(\alpha^*\alpha_*M))$ ([Lean 4: `Section3_2.unstableHeightNUnitCobarIso (UnstableChangeOfRings.lean)`]). Applying the Generalized Shapiro's Lemma (Proposition 2.14) to $N = \alpha_*M$ identifies the resolution $\mathrm{Hom}_{\mathcal{C}^{U_\Gamma}}(P, \alpha^*C^\bullet_{U_\Sigma}(\alpha_*M))$ with $\mathrm{Hom}_{\mathcal{D}^{U_\Sigma}}(\alpha_*P, C^\bullet_{U_\Sigma}(\alpha_*M))$. □

4. Formalization in Lean 4

The entire categorical development in Sections 2 and 3 has been formalized in the Lean 4 theorem prover [5] on top of Mathlib4 [11] in the package `ThesisLean` (comprising 2,533 lines of code across three modules with zero `sorry`s and zero compiler warnings).

4.1. **Mathematical Insights Gained from the Formalization.** Translating the thesis into Lean 4 led to several mathematical refinements and generalizations:

(1) **Dropping the Split-Localization Hypothesis in §3.2:** In the original thesis exposition, the comonad structure on $U_\Sigma(N) = K(n)_* \otimes_{BP_*} U_\Gamma(N)$ and the comonad functor $\alpha : K(n)_* \otimes_{BP_*} U_\Gamma(-) \to U_\Sigma(K(n)_* \otimes_{BP_*} -)$ used the algebraic simplification $K(n)_* \otimes_{BP_*} N = N$. In Lean 4, distinguishing $N \in \mathcal{D}$ from its restriction of scalars $U(N) \in \mathcal{C}$ revealed that $U_\Sigma := F \circ U_\Gamma \circ U$ is a comonad and $(F, FU_\Gamma(\eta))$ is a comonad functor for *any* comonad U_Γ and *any* adjunction $F \dashv U$, without assuming that the counit $\varepsilon : FU \to 1_\mathcal{D}$ is an isomorphism.

(2) **Definitional Equality on Cofree Coalgebras:** In the proof of Theorem 3.7 ($\alpha_*\alpha^* N \cong N$), the pushforward of a cofree U_Γ-coalgebra on UY satisfies $\alpha_*(\mathrm{cofree}_{U_\Gamma}(UY)) = \mathrm{cofree}_{U_\Sigma}(Y)$ *by definitional equality* (`rfl`) in Lean 4 (`unstablePushforward_obj_cofree`). This allowed us to prove that the comparison isomorphism $\alpha_*\alpha^* N \xrightarrow{\sim} N$ is literally the counit of the lifted adjunction $\alpha_* \dashv \alpha^*$ (`unstableCanonicalEqualizer_lift_eq_li`

(3) **Explicit Cobar Contractibility:** Rather than axiomatizing Bousfield's cosimplicial model structure [4], we formalized all five cosimplicial identities for the cobar complex $C^\bullet_G(X)$ and constructed the explicit left contracting homotopy on $\mathrm{Hom}_{\mathcal{C}^G}(C^\bullet_G(X), G(J))$ and $\mathrm{Hom}_{\mathcal{C}^G}(\alpha^* C^\bullet_H(N), G(J))$.

4.2. **Concordance of Paper Results and Lean 4 Declarations.** Table 1 lists the correspondence between the definitions, propositions, and theorems in this paper and their machine-checked declarations in `ThesisLean`.

Table 1: Concordance between mathematical results and Lean 4 declarations.

Paper Reference	Module	Lean 4 Declaration
Prop. 2.3 (i, j on $\mathcal{C}_G \leftrightarrows \mathcal{C}^G$)	`ComonadMorphisms`	`Section2_1.cokleisliToCoalgebraFullyFaithful`
Def. 2.4 (Comonad functor)	`ComonadMorphisms`	`Section2_2.ComonadFunctor`
Prop. 2.5 (Pushforward α_*)	`ComonadMorphisms`	`Section2_2.ComonadFunctor.pushforward`
Def. 2.6 (Comonad opfunctor)	`ComonadMorphisms`	`Section2_2.ComonadOpfunctor`
Prop. 2.7 (Kleisli pushforward β_*)	`ComonadMorphisms`	`Section2_2.ComonadOpfunctor.cokleisliPushforward`
Prop. 2.8 (Coalgebra pullback α^*)	`ComonadMorphisms`	`Section2_2.ComonadOpfunctor.coalgebraPullback`
Prop. 2.9(1) (Fundamental mate relation)	`ComonadMorphisms`	`Section2_3.mate_fundamental_rel_counit`

Paper Reference	**Module**	**Lean 4 Declaration**
Prop. 2.9(2) (Mate opfunctor (U, β))	`ComonadMorphisms`	`Section2_3.mateOpfunctor`
Prop. 2.10 (Lifted adjunction $\alpha_* \dashv \alpha^*$)	`ComonadMorphisms`	`Section2_3.liftedAdjunction`
Def. 2.11 (Induced monad & cobar $C^\bullet_G$)	`CobarResolution`	`Section2_4.inducedMonadOnCoalgebras`, `cobarObj`
Prop. 2.12(1) (Cosimplicial laws)	`CobarResolution`	`Section2_4.cobarCoface_comp_cobarCoface`
Prop. 2.12(2) (Left contractibility)	`CobarResolution`	`Section2_4.cobarHomLeftContractibleTower`
Prop. 2.13 (Shapiro triangle)	`CobarResolution`	`Section2_5.shapiro_naturality_triangle_coaug`
Prop. 2.14(1) (Weak G-resolution $\alpha^* C^\bullet_H$)	`CobarResolution`	`Section2_5.pullbackCobarCofreeLeftContractibleTower`
Prop. 2.14(2) (Generalized Shapiro's Lemma)	`CobarResolution`	`Section2_5.generalizedShapiroExtCobarIso`
Prop. 3.1 (Canonical equalizer)	`UnstableChangeOfRings`	`Section3_1.coalgebraCanonicalEqualizerIso`
Stable Change of Rings (§3.1)	`UnstableChangeOfRings`	`Section3_1.stableChangeOfRingsCobarIso`
Def. 3.2 & Thm. 3.3 (U_Σ comonad)	`UnstableChangeOfRings`	`Section3_2.unstableSigmaComonad`
Def. 3.4 & Thm. 3.5 (α morphism)	`UnstableChangeOfRings`	`Section3_2.unstableComonadFunctor`
Thm. 3.6(1) ($\beta_N = \eta_{U_\Gamma U N}$)	`UnstableChangeOfRings`	`Section3_2.unstableMateNatTrans_app`
Thm. 3.6(2) ($\alpha_* \dashv \alpha^*$ for U_Σ)	`UnstableChangeOfRings`	`Section3_2.unstableLiftedAdjunction`
Thm. 3.7 ($\alpha_* \alpha^* N \cong N$)	`UnstableChangeOfRings`	`Section3_2.unstablePushforwardCotensorIso`, `isIso_unstableLiftedCounit`
Prop. 3.8(1–2) (Excess bounds)	`UnstableChangeOfRings`	`Section3_2.excess_vn_shift_iff`, `excess_ul_action_iff`
Prop. 3.8(3) (v_n^{-1} & colimits)	`UnstableChangeOfRings`	`Section3_2.prop3_2_2_colimit_iso`
Prop. 3.9(1) ($\alpha^* U_\Sigma(Y) \cong U_\Gamma(UY)$)	`UnstableChangeOfRings`	`Section3_2.prop3_2_4_cotensor_cofree`
Prop. 3.9(2) ($\alpha^* \alpha_* M \cong B(n)_* \otimes M$)	`UnstableChangeOfRings`	`Section3_2.prop3_2_3_unitPullbackIso`
Prop. 3.9(3) ($M \xrightarrow{\sim} \alpha^* \alpha_* M$)	`UnstableChangeOfRings`	`Section3_2.liftedUnitIsoOfHeightNConnected`
Thm. 3.10 (Unstable Change of Rings)	`UnstableChangeOfRings`	`Section3_2.unstableChangeOfRingsCobarIso`

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